

Incident Report Analysis

DoS Attack via ICMP Flood | NIST Cybersecurity Framework Application

SUMMARY

The organization's internal network went down for approximately two hours after a malicious actor flooded it with ICMP packets through a firewall that hadn't been properly configured. Because there was no rule in place to filter or limit that type of traffic, the flood passed through unchecked and overwhelmed network services until they stopped responding entirely.

Once the attack was detected, the incident response team moved quickly: incoming ICMP traffic was blocked at the firewall, non-critical services were taken offline to reduce internal load, and critical services were restored as fast as possible. After the immediate threat was contained, the team put controls in place to prevent the same thing from happening again.

IDENTIFY

The team ran an audit of the network infrastructure after the attack and quickly identified the root cause: the firewall had no rule to filter or rate-limit incoming ICMP traffic. That single misconfiguration was all the attacker needed. Every system connected to the internal network was affected for the duration of the outage, and no critical data appeared to have been exfiltrated since this was a volumetric availability attack, not an intrusion.

The audit also flagged that there was no active network monitoring in place before the attack, which meant there was no automated detection and the response was delayed. That gap was noted for immediate remediation.

PROTECT

To close the door on this specific attack vector, a new firewall rule was implemented to cap the rate of incoming ICMP packets so no single source can saturate the network. Source IP verification was also enabled on the firewall to flag and drop packets with spoofed addresses, which is a common tactic in ICMP-based flooding attacks.

An IDS/IPS system was deployed to continuously scan ICMP traffic for suspicious patterns and automatically filter traffic that matches known attack signatures. Network access controls were reviewed across the board and updated to reduce the overall attack surface.

DETECT

Network monitoring software is now running and establishing traffic baselines so the team has visibility into what normal looks like. Anything outside that baseline, including sudden spikes in ICMP volume, will trigger an alert before it has a chance to cause an outage.

The IDS/IPS system adds another detection layer by checking traffic in real time against known attack signatures. Together, these tools cut response time significantly compared to where the organization was before this incident.

RESPOND

As soon as the flood was identified, the team applied a firewall rule to block all incoming ICMP packets, stopping the attack at the perimeter. Non-critical network services were taken offline to take pressure off the network, while critical services were kept running where possible. The response was handled internally before escalating outward.

Upper management was briefed on the incident and the actions taken. An internal incident report was filed documenting the timeline, what failed, and how the team responded. No customer data was affected, so external notification was not required, but the event was logged per organizational policy.

RECOVER

Recovery was staged intentionally. Critical services came back first once the firewall rule was confirmed to be holding. Non-critical services followed in sequence after each was verified to be stable. No full restoration happened until the firewall config was reviewed and locked down to prevent the same vulnerability from being immediately re-exploited.

All services were fully restored within the two-hour window. The team was debriefed afterward and walked through the new controls now in place. The incident surfaced a real gap in the organization's monitoring posture and the response accelerated fixes that should have been in place from the start.